COURSE TITLE : PROCESS CONTROL AND INSTRUMENTATION
COURSE CODE : 5095
COURSE CATEGORY : E
PERIODS/WEEK : 4
PERIODS/SEMESTER: 52/5
CREDITS : 4

TIME SCHEDULE

Module	Topics	Period
1	General characteristics and Temperature measuring instruments	13
2	Pressure measuring instruments	13
3	Flow and level measuring instruments	13
4	Controllers, Hydraulic systems and Hydraulic pumps	13
TOTAL		**52**

SPECIFIC OUTCOME

MODULE I GENERAL CHARACTERISTICS AND TEMPERATURE MEASURING INSTRUMENTS

1.1.0 To understand the characteristics of instruments for measuring process parameters

1.1.1. List the static and dynamic characteristics of measuring instruments.
1.1.2. Define Calibration,accuracy,Precision, sensitivity, Repeatability,Reproducibility,Fidelity and Lag.

1.2.0 To know the temperature measuring instruments.

1.2.1. Discuss the various temperature scales and the conversion of Celsius scale into Fahrenheit scale with illustration.
1.2.2. List the various temperature measuring instruments.
1.2.3. Explain the principle,working,construction,merits and demerits of Mercury in glass and metal thermometer, Filled system thermometer, Bimetallic thermometer and Thermocouples.
1.2.4. Explain the principle,working,construction,merits and demerits of Optical and Radiation Pyrometers

MODULE II PRESSURE MEASURING INSTRUMENTS

2..1.0 To know the Pressure measuring instruments and its applications.

2.1.1. Define Absolute, Guage,Atmospheric and Vacuum Pressures.
2.1.2. List the Types of Pressure Tranducers.
2.1.3. Explain the principle, working,construction, advantages and disadvantages of U- tube manometer,Well type and Micromanometer.
2.1.4. Describe the principle,working,construction,advantages and disadvantages of C-bourdon tube,Diaphragms and Bellows. .

2.1.5. Explain the principle,working,construction, merits and demerits of Strain Guages and Piezoelectric pressure transducer.
2.1.6. Explain the principle,working,construction,advantages and disadvantages of L.V.D.T.

<u>MODULE III</u> FLOW AND LEVEL MEASURING INSTRUMENTS

3.1.0 To understand the flow and level measuring instruments and its applications.

3.1.1. Differentiate between rate of flow and total flow.
3.3.2. State Bernoull's Theorem.
3.1.3. Explain the principle,working,construction,advantages and disadvantages of Venturimeter and Orifice meter.
3.3.4. Explain the principle,working,advantages and disadvantages of Variable area meter (Rotameter).
3.1.5. Describe the principle,working,advantages and disadvantages of Magnetic flow meter,Turbine meter and Lobed impeller flow meter.
3.3.6. Explain the principle,working,construction,advantages and disadvantages of
3.1.7. List the direct and indirect level measuring instruments.
3.1.8. Explain the principle,working,construction,advantages and disadvantages of Hook type level indicator,sight glass and Float type level indicator.
3.1.9. Explain the principle,working ,advantages and disadvantages of Air purge system(Bubbler type) of level measurement.

<u>MODULE IV</u> CONTROLLERS, HYDRAULIC SYSTEMS AND HYDRAULIC PUMPS

4.1.0 To understand the types of controllers and its applications in polymer industries ,the Hydraulic systems and hydraulic pumps.

4.1.1. List the types of control systems used in industries.
4.1.2. Differntiate between open loop and closed loop control systems with examples.
4.1.3. Explain the Pneumatic and Electronic Controllers.
4.1.4. Describe the ON-OFF,Proportional, derivative and integral control actions.
4.4.5. Compare the characteristics of ON-OFF,Proportional, derivative and integral control actions.
4.1.6. Explain P.I.D. controllers.
4.1.7. Describe the temperature and pressure control systems in moulding,extrusion and calendaring processes.
4.1.8. Explain the principle ,different parts and working of Hydraulic Press.
4.1.9. Explain the principle ,different parts and working of Hydraulic Lift and Hydraulic Crane.
4.1.10. Describe the working of centrifugal pump,Reciprocating pump,Jet pump,Vane pump and Rotary pump.
4.1.11. Compare centrifugal pump with Reciprocating pump

CONTENT DETAILS

MODULE-I

General characteristics and Temperature measuring instruments

Static characteristics-Calibration, accuracy, precision, sensitivity, repeatability, reproducibility.Dynamic characteristics-Fidelity and lag, Temperature scales-Celsius, Fahrenheit and Kelvin,Temperature measuring instruments- Bimetallic thermometer, Mercury in glass thermometer, mercury in metal thermometer, Filled system thermometer. Resistance thermometer- Wheatstone bridge circuit, Thermocouples-see beck effect, Pyrometers-Optical and Radiation pyrometers.

MODULE-II

Pressure measuring instruments

Different pressures-Absolute, Guage, Atmospheric and Vacuum Pressures, Manometer-U-tube ,Well type and Micromanometer. Elastic pressure transducers-C bourdon tube, Diaphragms and Bellows. Electrical pressure transducers-Strain gauges, piezoelectric pressure transducers and LVDT

MODULE-III

Flow and Level measuring instruments

Rate flow-Total flow, Bernoulli's theorem- venture meter and Orifice meter Variable area meter, Magnetic flow meter, Turbine flow meter and lobed impeller flow meter Direct level measuring instruments-hook type level indicator, sight glass and Float type indicator. Indirect level measuring instruments-Air purge system

MODULE-IV

Controllers hydraulic system and hydraulic pumps

Types of control systems –open loop and closed loop control systems ,pneumatic and electronic controllers .Control action –ON –OFF , Proportional , Derivative and Integral control actions and PID controllers .Temperature and Pressure –Measurement and control –Molding _Extrusion and Calendaring process . Principle, Parts and Working –Hydraulic press , hydraulic Lift and Hydraulic Crane. Working - Centrifugal pump , Reciprocating pump , Jet pump , vane pump and rotary pump .

REFERENCE BOOKS

1. A Course in Mechanical measurment and Instrumentation- A.K.Sawhney (Publisher: Dhanpat Rai, New Delhi Edition: 12
2. Industrial Instrumentation -Donald P Echman(Publisher Wiley 1951)
3. Instrumentation -Franklyn.W.Kirk & Nicholas R.Rimboi (Publisher – American Technical Society1962)
4. Textbook of Hydraulics, Fluid Mechanics and Hydraulic Machines - R.S.Khurmi (Publisher- S. Chand Limited, 1987)